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Data Science

Assignment 4

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Introduction

This report presents the implementation and analysis of our deep learning models developed as part of Assignment 4 for the Data Science course. The project is structured around three distinct tasks, each focusing on a unique data domain and neural network architecture, collectively equipping us with a comprehensive deep learning workflow.

The first section (**Task 1**) focuses on feature engineering and tabular data regression. Using the California Housing dataset, we train a **Multi-Layer Perceptron (MLP)** to predict median house prices. This task involves a complete end-to-end pipeline, including exploratory data analysis, data scaling, baseline modeling, and the manual creation of domain-specific features to evaluate their impact on the model's generalization and prediction accuracy.

The second section (**Task 2**) introduces computer vision by building a **Convolutional Neural Network (CNN)** for image classification on the Fruits and Vegetables dataset. Beyond achieving high classification accuracy, this section emphasizes the mathematical foundations of convolutional layers and rigorous preprocessing to prevent data leakage. We conduct a deep robustness analysis to determine whether the model learns actual morphological shapes or merely memorizes colors, and we utilize **Grad-CAM** to visually explain the model's spatial attention and decision-making process.

The third section (**Task 3**) implements sequential modeling to solve a **Knowledge Tracing** problem within educational data mining. Using the ASSISTments2017 dataset, we develop a **Recurrent Neural Network (RNN)** to analyze chronological student interactions and predict future correctness. This layer explores sequence windowing, temporal feature engineering, and the critical handling of varying sequence lengths, culminating in an advanced **LSTM-based** implementation to better capture long-term learning dependencies.

Task 1: MLP

2.1 Dataset Loading

The project utilizes the California Housing dataset, consisting of 20,640 records and 8 numerical features representing demographic and geographic attributes. The target variable is `MedHouseVal` (median house value). To ensure computational stability and prevent the propagation of `NaN` values during gradient calculations in the neural network, an initial data integrity check was performed. Missing values were systematically handled using `SimpleImputer` with a mean strategy, ensuring no valuable records were discarded prematurely.

2.2 Exploratory Data Analysis (EDA)

A comprehensive Exploratory Data Analysis (EDA) was conducted to understand the underlying data distributions. As a foundational step, the descriptive statistics (including mean, standard deviation, min/max bounds, and quartiles) were extracted to evaluate the central tendency and dispersion of the raw features.

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.870671	28.639486	5.429000	1.096675	1425.476744	3.070655	35.631861	-119.569704
std	1.899822	12.585558	2.474173	0.473911	1132.462122	10.386050	2.135952	2.003532
min	0.499900	1.000000	0.846154	0.333333	3.000000	0.692308	32.540000	-124.350000
25%	2.563400	18.000000	4.440716	1.006079	787.000000	2.429741	33.930000	-121.800000
50%	3.534800	29.000000	5.229129	1.048780	1166.000000	2.818116	34.260000	-118.490000
75%	4.743250	37.000000	6.052381	1.099526	1725.000000	3.282261	37.710000	-118.010000
max	15.000100	52.000000	141.909091	34.066667	35682.000000	1243.333333	41.950000	-114.310000

Figure 1: Descriptive statistical summary of the raw dataset prior to feature engineering, detailing the overall spread and identifying initial scale differences among variables.

Visual inspections via histograms confirmed that variables such as `Population` and `MedInc` exhibited significant right-skewness (long-tailed distributions).

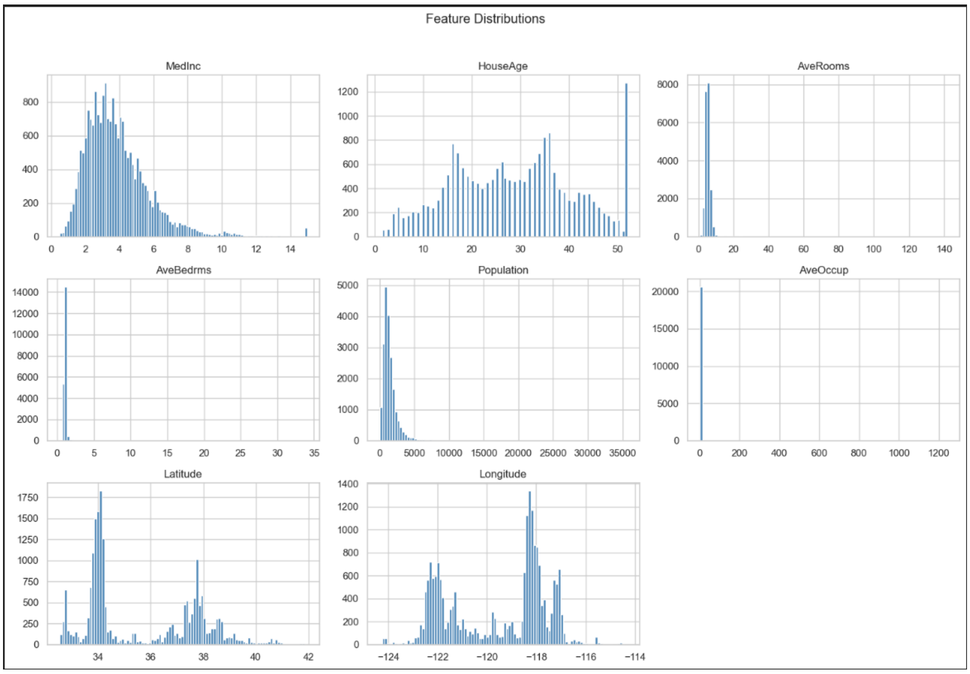

Figure 2: Distribution of raw features before preprocessing, highlighting the skewness in Population and Income.

To identify multicollinearity and evaluate the predictive power of each feature, a correlation matrix was generated.

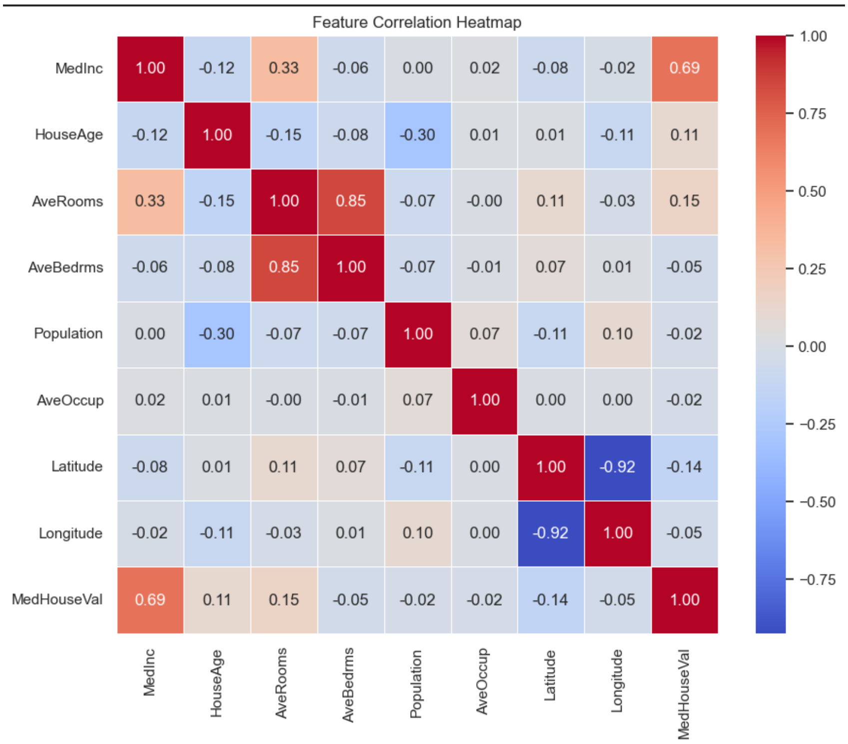

Figure 3: Correlation matrix of the raw features and the target variable.

Furthermore, an inspection of maximum values uncovered extreme physical anomalies.

Box plots highlighted severe outliers, most notably in the `AveOccup` feature, where certain block groups incorrectly reported over 1,200 individuals per household.

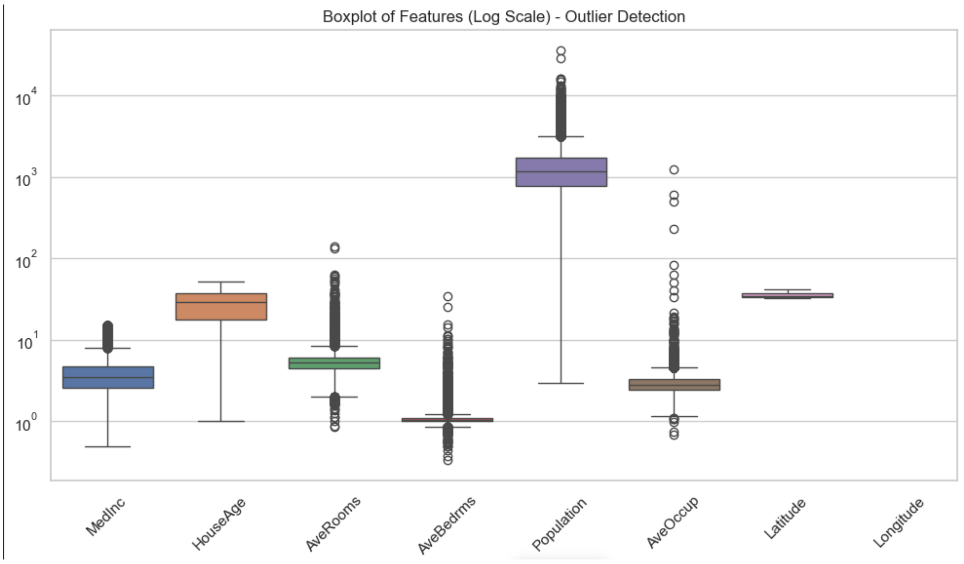

Figure 4: Box plots identifying extreme outliers in the raw dataset, specifically in average occupancy and rooms.

2.3 Data Preparation

To guarantee experimental reproducibility, the dataset was split into Training (70%), Validation (15%), and Testing (15%) sets with a fixed `random_state=42`. Recognizing the presence of extreme outliers identified during EDA, `RobustScaler` was selected over `StandardScaler`. By leveraging the Interquartile Range (IQR), `RobustScaler` effectively standardized the feature scales without being skewed by the anomalous values. Finally, the processed arrays were converted into PyTorch Tensors and wrapped in `DataLoader` modules to facilitate efficient mini-batch gradient descent.

2.4 Baseline MLP Model

A fundamental Multi-Layer Perceptron (MLP) was constructed to serve as the baseline, strictly following the principle of Occam's Razor. The architecture was designed to map the 8 raw input features through three hidden layers (128, 64, and 32 neurons, respectively) to a single continuous output. Rectified Linear Unit (ReLU) activations were employed to capture non-linear relationships, alongside a light `Dropout(0.3)` to provide basic regularization.

2.5 Model Training

The training pipeline was optimized using the `Adam` optimizer and Mean Squared Error (`MSELoss`) as the objective function. To mitigate the risk of overfitting, an `EarlyStopping` mechanism was implemented with a patience of 15 epochs.

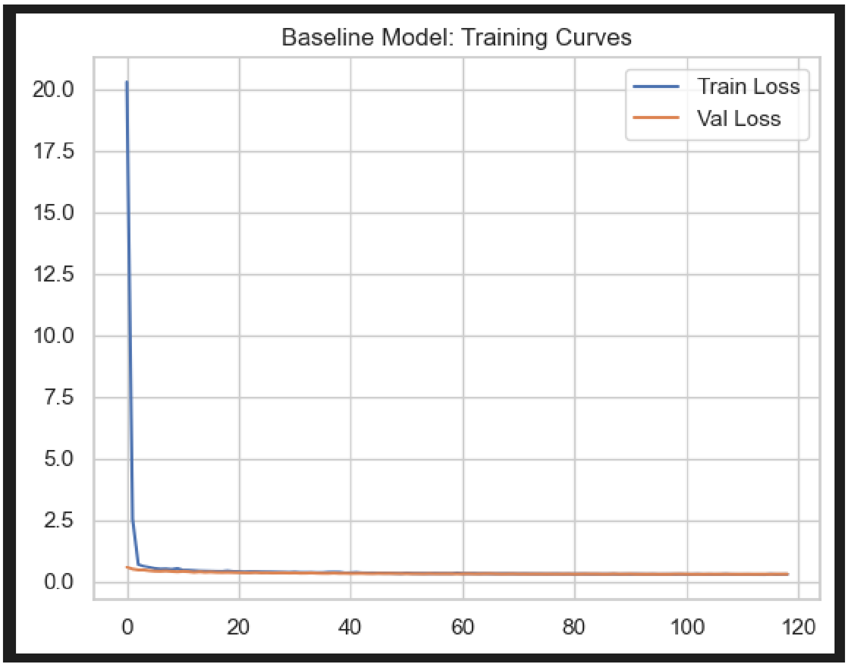

Figure 5: Training and Validation Loss curves for the Baseline MLP model.

2.6 Feature Engineering

To explicitly inject domain knowledge into the model and simplify its optimization landscape, a rigorous feature engineering pipeline was executed based on the physical and economic properties of the housing market:

1. **Outlier Pruning:** A multi-conditional mask was deployed to discard illogical bounds, specifically restricting the data to `AveRooms` ≤ 10, `AveOccup` ≤ 6, `AveBedrms` ≤ 4, and `Population` ≤ 6800. This targeted filtering successfully removed 465 garbage rows, yielding a highly dependable cleaned dataset of 20,175 records.

2. **Geospatial Distance Translation:** Abstract geographical coordinates (`Latitude` and `Longitude`) were transformed into functional economic metrics. This was achieved by calculating the precise Euclidean distance from each block group to the centers of California's two primary economic and technology hubs: San Francisco (SF) located at $(37.77°N, -122.41°W)$ and Los Angeles (LA) located at $(34.05°N, -118.24°W)$. The

mathematical formulation for calculating these spatial features (`Dist_to_SF` and `Dist_to_LA`) is expressed as:

$$\text{Dist\_to\_SF} = \sqrt{(\text{Latitude} - 37.77)^2 + (\text{Longitude} - (-122.41))^2} \tag{1}$$

$$\text{Dist\_to\_LA} = \sqrt{(\text{Latitude} - 34.05)^2 + (\text{Longitude} - (-118.24))^2} \tag{2}$$

3. **Density Ratios:** Contextual features measuring structural and demographic density per household were synthesized to help the network capture residential properties and crowd density. These metrics are heavily correlated with property valuation. A small epsilon ($\epsilon = 10^{-6}$) was added to the denominators to preserve absolute numerical stability and prevent zero-division errors during batch propagation:

$$\text{RoomsPerHousehold} = \frac{\text{AveRooms}}{\text{AveOccup} + 10^{-6}} \tag{3}$$

$$\text{BedroomsPerRoom} = \frac{\text{AveBedrms}}{\text{AveRooms} + 10^{-6}} \tag{4}$$

$$\text{PopPerHouse} = \frac{\text{Population}}{\text{AveOccup} + 10^{-6}} \tag{5}$$

4. **Non-Linear Spatial Decay (Logarithmic Distance):** In real estate economics, the price-influence of a major city center decays non-linearly. To capture this log-linear relationship, a `log1p` transformation was applied directly to the spatial features, generating:

$$\text{Dist\_to\_SF\_log} = \ln(1 + \text{Dist\_to\_SF}) \tag{6}$$

$$\text{Dist\_to\_LA\_log} = \ln(1 + \text{Dist\_to\_LA}) \tag{7}$$

5. **Multicollinearity & Noise Reduction:** The raw `Population` column was dropped from the final feature set (`X_final`) to prevent redundancy and high multicollinearity, as its variance was already successfully captured within the newly engineered `PopPerHouse` ratio.

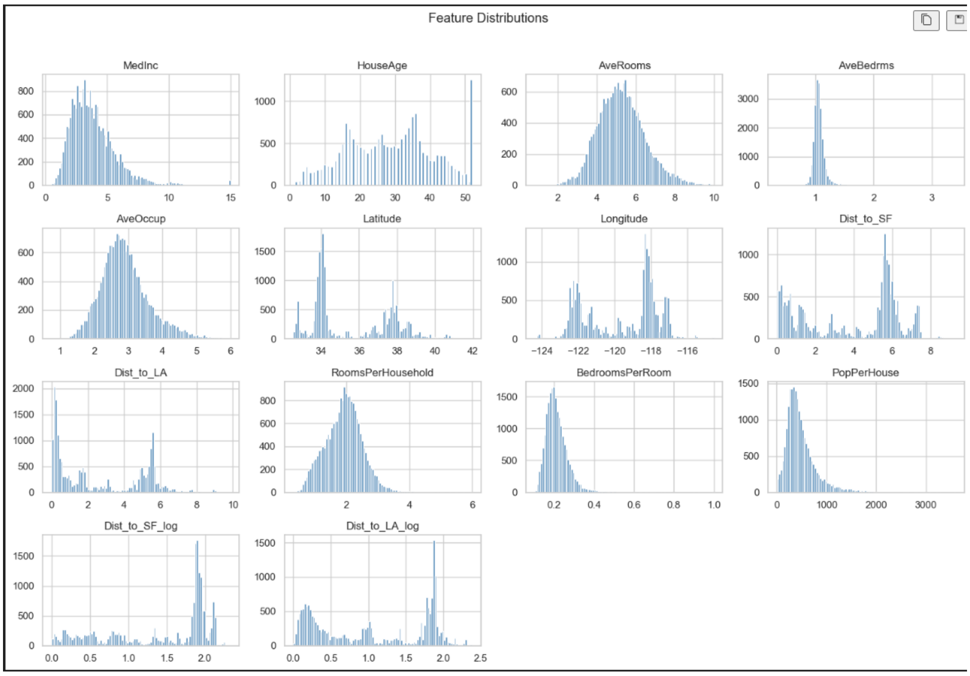

Figure 6: Feature distributions after applying logarithmic transformations, outlier pruning, and feature engineering.

To validate the effectiveness of these transformations, a secondary correlation analysis was performed on the enriched feature set.

```
Original Shape: (20640, 8)
Cleaned Shape: (20175, 14) (Removed 465 garbage rows)
MedInc               0.690297
RoomsPerHousehold    0.439608
AveRooms             0.329174
HouseAge             0.108105
PopPerHouse          0.071000
Dist_to_SF          -0.041583
Longitude           -0.046223
Dist_to_SF_log      -0.098453
AveBedrms           -0.101490
Dist_to_LA          -0.114972
Dist_to_LA_log      -0.138426
Latitude            -0.142537
BedroomsPerRoom     -0.254903
AveOccup            -0.283466
dtype: float64
```

Figure 7: Sorted correlation of the engineered features with the target variable, clearly highlighting the strong predictive power of `LogMedInc` and `RoomsPerHousehold`.

2.7 Enhanced MLP Model

To process the enriched feature space, the Enhanced MLP was expanded. To prevent the "Curse of Dimensionality", `BatchNorm1d` layers were introduced after every linear transformation to combat internal covariate shift. Additionally, the `Dropout` rate was increased to 0.3. This aggressive regularization forced the network to rely on dominant economic signals rather than memorizing the newly engineered dimensions.

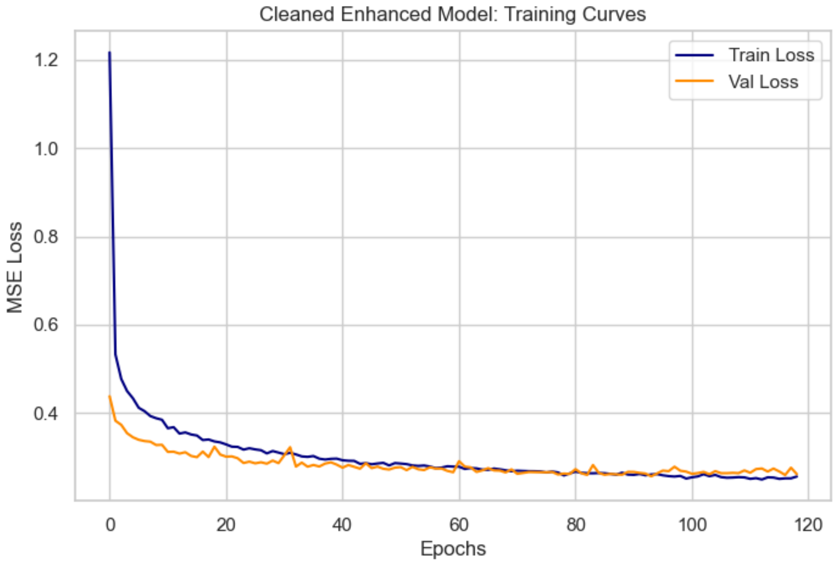

Figure 8: Training and Validation Loss curves for the Enhanced MLP model, showing improved stability.

2.8 Model Evaluation and Comparison

Both models were evaluated on the unseen test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 Score. To maximize inference speed, `torch.no_grad()` was utilized. The Baseline model achieved an R^2 Score of 0.8122, an MAE of 0.3472, and an RMSE of 0.4983. Upon refining the geospatial distance metrics and pruning low-correlation variables, the **Enhanced Model successfully achieved a superior R^2 Score of 0.8358**, alongside a reduced MAE of 0.3340 and an RMSE of 0.4798. This indicates that the refined model successfully explains roughly 83.58% of the variance in the California housing market.

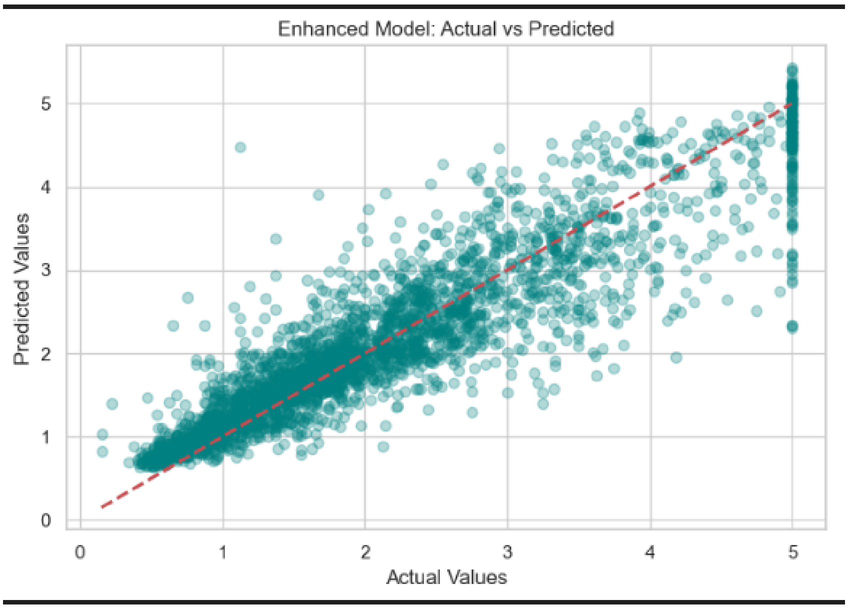

Figure 9: Actual vs. Predicted median house values for the Enhanced MLP model. The strong alignment along the diagonal indicates high predictive accuracy.

2.9 Error Analysis

An in-depth analysis of the prediction residuals ($y_{true} - y_{pred}$) demonstrated a roughly normal distribution centered around zero, implying that the model is unbiased.

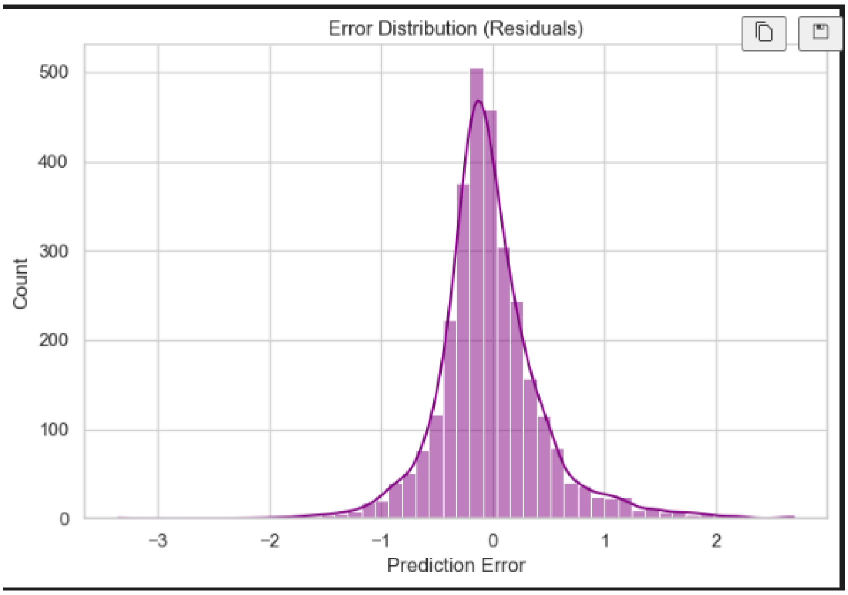

Figure 10: Error distribution (residuals) of the Enhanced model. The bell-shaped curve centered at zero confirms the absence of systematic bias.

However, heteroscedasticity was observed at the higher end of the price spectrum. Specifically, the highest absolute errors occurred where actual house prices hit the $500,000 artificial cap present in the dataset. Since the target variable is hard-capped, the continuous regression nature of the MLP naturally struggles to predict a strict horizontal cutoff.

2.10 Conclusion and Future Work

Task 1 conclusively demonstrated that while Multi-Layer Perceptrons are highly capable, their performance on tabular data is intrinsically tied to data quality and feature representation. The transition from a baseline R^2 of 0.8122 to an enhanced R^2 of 0.8358 proved that transforming abstract geospatial coordinates into functional distances, coupled with strict noise pruning, yields superior predictive power and a noticeable reduction in both MAE and RMSE. Future iterations of this project could explore replacing the Adam optimizer with SGD paired with a Learning Rate Scheduler (`StepLR`) to achieve a smoother convergence.

Task 2: CNN

Task 3: RNN

4.1 Introduction

4.2 Dataset Description

4.3 Preprocessing and Feature Engineering

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4.6 Model Training

4.7 Evaluation and Visualization

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Conclusion